

CHARMAINE GUO

Wayland, MA 01778 & Evanston, IL 60201
(407) · 820 · 7755 ◊ guo.charmaine.cg@gmail.com
Website: charmaineguo.com

EDUCATION

Northwestern University
B.S. in Industrial Engineering

June 2028

SKILLS

Computer Languages
Soft Skills
Hard Skills
Organizational Tools

Python, Java, MATLAB, HTML, CSS
Project Management, Teamwork, Time-Management, Adaptability
Research Methods, Data Collection, 3D Printing, SolidWorks
Microsoft Excel, Google Drive

EXPERIENCE

Student Tech Aide
The Garage

September 2025 – Present
Evanston, IL

- Manage makerspace operations and inventory for 3D printers, CNC, and woodworking tools, utilizing data-informed logistics to ensure equipment uptime and optimal material stock levels.
- Facilitate resident fabrication by translating complex project requirements into technical workflows, providing expert guidance on additive and subtractive manufacturing techniques and safety.

Undergraduate Researcher — Schuh Group
Northwestern University

Jan 2026 - March 2026
Evanston, IL

- Synthesizing zircon (ZrSiO₄) coatings on micron-scale sand particles using the Pechini method, leveraging polymerizable precursors to ensure precise stoichiometric control and chemical homogeneity.
- Refining sol-gel processing routes by adjusting citric acid to ethylene glycol ratios, improving the stability of the resin precursor for more consistent particle encapsulation.

Quality Intern
Littelfuse

June 2025 - August 2025
Beverly, MA

- Engineered a centralized data repository using Power Query to automate the ETL process for 300+ legacy quality reports, standardizing disparate data formats into a relational structure.
- Architected a multi-parameter search engine by indexing part numbers, package types, and leadframe variables, enabling instantaneous data retrieval and cross-functional visibility.
- Optimized data retrieval latency, reducing mean search time from several hours to under 60 seconds for the Beverly engineering team.

PROJECTS

Purrfect Care
Python, PyQt5, Gemini API

Desktop (WildHacks)
March 2025

- Built a PyQt5 desktop app with wellness reminders, to-do list, and animated notifications.
- Integrated Gemini API for contextual chat with a virtual assistant cat.

Skynet System
Onshape, Arduino IDE, ESP32

Team Project (Design Thinking & Communication)
January - March 2025

- Designed solutions for remote zoo access, with 3d printed prototypes including an interactive online tour prototype built with web technologies and a Hexbug like design.
- Final deliverable: a zipline-mounted camera system enabling live video feed controlled remotely by users.

EXTRACURRICULAR ACTIVITIES

Low Voltage Team
Northwestern Formula SAE (NUFSAE)

October 2024 – Present

- Engineered the vehicle's Power Distribution Module (PDM) using Altium Designer, overseeing the full schematic capture and PCB layout process.
- Developing comprehensive wiring harness specifications, including connector selection, pinout mapping, and signal integrity routing.
- Leading technical onboarding for new members by delegating design tasks and conducting training sessions on RapidHarness and industry-standard documentation.